

BUAN 301

Session 3 : Descriptive Statistics and Summarization

(28th Aug, 2026)

Agenda

□ Review of Descriptive Statistics

- Location
- Variability
- Shape

□ Summarization of Structured Data using Excel

- Data Set : WTO_60_Countries

Location

□ "Around what values is the variable located?"

□ Measures of Centrality

- Mean - Average value of all observations

$$\mu = \frac{1}{n} \sum_i X_i$$

- Median - Centrally located value in the sorted list of observations
- Mode - Value/Category with the most frequent occurrence

□ Measure of location - percentiles/quartiles/deciles etc.

- The value at location

$$L_p = \frac{p}{100} (n + 1)$$

Variability (Range)

□ “What is the dispersion of the values”?

□ Range

- Difference between the largest and smallest values

□ Interquartile Range

- Difference between X_{Q_1} and X_{Q_3} where $Q_1 = \frac{n+1}{4}$ and $Q_3 = \frac{3(n+1)}{4}$

Variability (Variance)

□ “What is the dispersion of the values”?

□ Variance (Population)

- Average of the square of the deviation from the mean

$$\sigma^2 = \frac{1}{n} \sum_i (X_i - \bar{X})^2$$

□ Variance (Sample)

$$s^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$$

□ Standard Deviation is square root of Variance i.e σ or s

Shape

□ **Skewness** – Measures the degree of symmetricity around mean

$$= \frac{n}{(n-1)(n-2)} \sum \left(\frac{X_i - \bar{X}}{s} \right)^3$$

- Can be positive or negative depending upon the skew

□ **Kurtosis** – Measures the strength of the tail of the distribution

$$= \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{X_i - \bar{X}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

- Can be positive or negative relative to the normal distribution

References

- ❑ Statistics for Business & Economics, 13th Ed. (*Anderson/Sweeney/Williams/ Camm/ Cochran*)
 - Chapter 3 – Descriptive Statistics

- ❑ Business Analytics : The Science of Data-driven Decision Making (Wiley) - *U. Dinesh Kumar*
 - Chapter 2: Foundations of Data Science : Descriptive Analytics

Next Class

□ Topic for Discussion

- Probability Review

□ Pre-reads,

- Chapter 4 – Introduction to Probability. Statistics for Business & Economics, 13th Ed. (Anderson/Sweeney/Williams/ Camm/ Cochran)
- Chapters 1,2 – First Course in Probability, 10th Ed. (Sheldon Ross)

Thank You!

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